



Grade 8 General Science

Directions:

On the following pages of your practice test booklet are questions for the Grade 8 *Nebraska Student-Centered Assessment System (NSCAS) General Science* practice test.

Read these directions carefully before beginning the test.

This practice test will include several different types of questions. Multiple choice questions will ask you to select an answer from among several answer choices. Multiple select questions will ask you to select multiple correct answers from among five or more answer choices. For some questions, there may be two parts, Part A and Part B. Some questions will ask you to construct an answer by following the directions given.

For all questions:

- Read each question carefully and choose the best answer.
- Review each question carefully. You may not return to a previous question.
- You may use scratch paper to make notes.
- Be sure to answer ALL the questions.

1. Use the information to complete the task.

Seawatch Bay

People on an evening fishing trip in Seawatch Bay notice that the water faintly glows at night. The glow in the water is stronger around their boat as it moves through the bay. They collect a water sample and send it to a marine science lab for analysis.

Scientists examine the water sample under a microscope. They find a microorganism named *Noctiluca scintillans*. The *Noctiluca scintillans* in the water sample exhibits two different traits. Some of the microorganisms in this species can produce light, while others cannot.

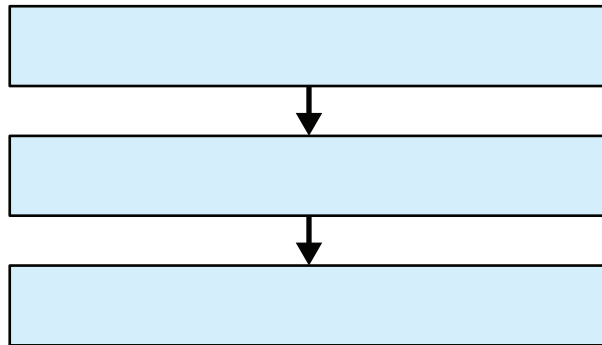
The *Noctiluca scintillans* that can produce light make a protein called luciferase. The luciferase protein helps drive a chemical reaction that releases light.

Write the phrases in the flow chart to describe the process that allows *Noctiluca scintillans* to produce light.

trait to produce light

luciferase gene

luciferase protein

Light Production in *Noctiluca scintillans*

Review each question carefully. You may not return to a previous question.

2. Use the information to answer the question.

Seawatch Bay

The *Noctiluca scintillans* in the water sample exhibits two different traits. Some of the microorganisms in this species can produce light, while others cannot.

Scientists at the marine science lab wonder if the *Noctiluca scintillans*' ability to produce light affects the probability of it surviving in Seawatch Bay.

What evidence should the scientists collect to determine whether the light-producing trait affects the *Noctiluca scintillans*' ability to survive in Seawatch Bay?

Select **all** that apply.

- A. whether prey species are attracted to the *Noctiluca scintillans* that are able to glow
- B. whether water pollutants limit the size of the *Noctiluca scintillans* population in Seawatch Bay
- C. whether *Noctiluca scintillans* populations in other bodies of water have the same light-producing traits
- D. whether the *Noctiluca scintillans* reproduce more quickly when the bay's water temperature is warmer
- E. whether predators have an easier time capturing *Noctiluca scintillans* with the light-producing trait

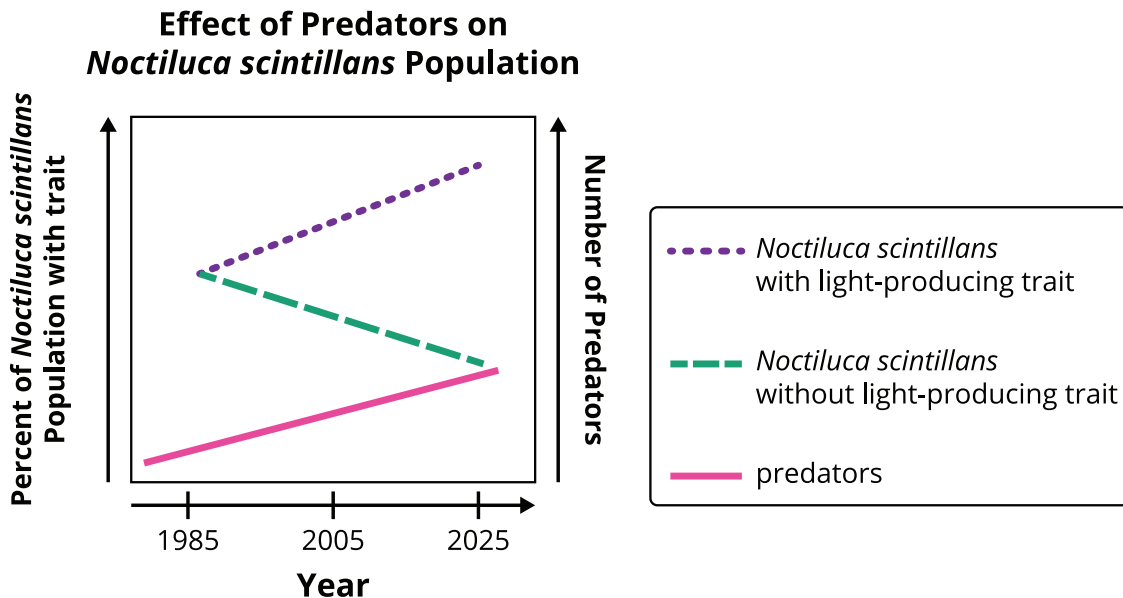
Review each question carefully. You may not return to a previous question.

3. Use the information to answer the question.

Seawatch Bay

Scientists at the marine science lab wonder if the *Noctiluca scintillans*' ability to produce light affects the probability of it surviving in Seawatch Bay.

The scientists collect data on how the number of *Noctiluca scintillans* in Seawatch Bay with the light-producing trait and without the light-producing trait has changed in relation to the predator population that eats them. The graph shows the data.



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Which statement BEST describes the effect of predators on the *Noctiluca scintillans* population over time?

- A. As the percentage of *Noctiluca scintillans* that produce light increased, the population of predators remained the same.
- B. As the percentage of *Noctiluca scintillans* that are not able to produce light increased, the population of predators decreased.
- C. As the population of predators increased, the percentage of *Noctiluca scintillans* that produce light increased in the population.
- D. As the population of predators decreased, the percentage of *Noctiluca scintillans* that produce light remained the same in the population.

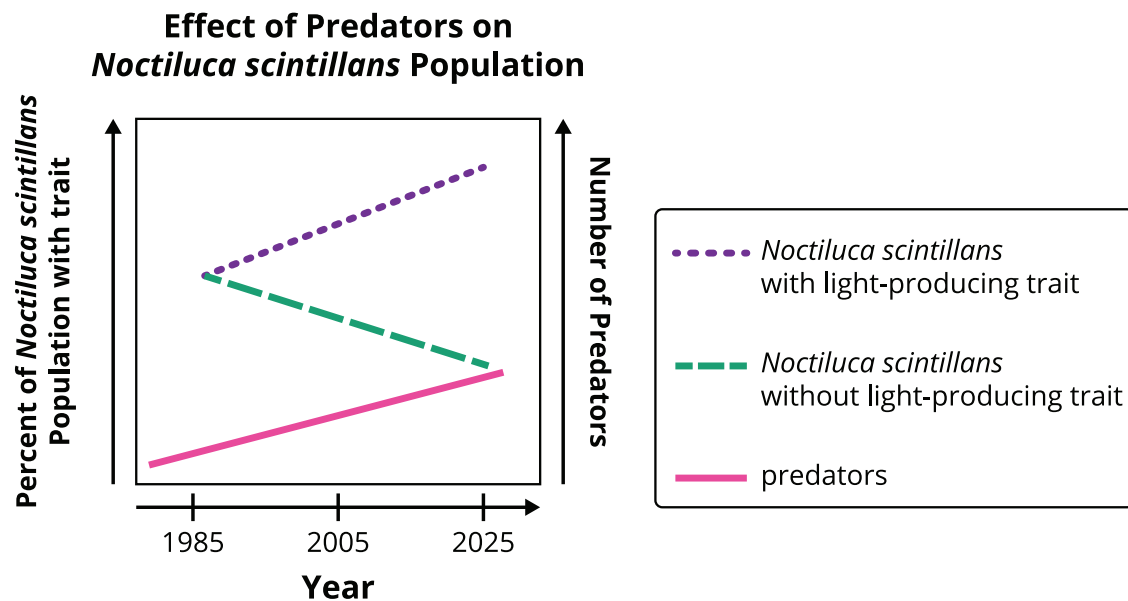
Review each question carefully. You may not return to a previous question.

4. Use the information to answer the question.

Seawatch Bay

Scientists at the marine science lab wonder if the *Noctiluca scintillans*' ability to produce light affects the probability of it surviving in Seawatch Bay.

The scientists collect data on how the number of *Noctiluca scintillans* in Seawatch Bay with the light-producing trait and without the light-producing trait has changed in relation to the predator population that eats them. The graph shows the data.



This question has two parts. Answer Part A, then answer Part B.

Part A

Which statement BEST explains why the percentage of *Noctiluca scintillans* that produce light in the population is increasing?

- A. *Noctiluca scintillans* that produce light attract predators.
- B. The ability to produce light helps the *Noctiluca scintillans* avoid predators.
- C. The ability to produce light decreases the likelihood that the *Noctiluca scintillans* will reproduce.
- D. *Noctiluca scintillans* that produce light are less likely to survive than those that do not produce light.

Part B

Which claim about the Seawatch Bay *Noctiluca scintillans*' light-producing trait is supported by evidence?

Select the correct choices to complete the sentence.

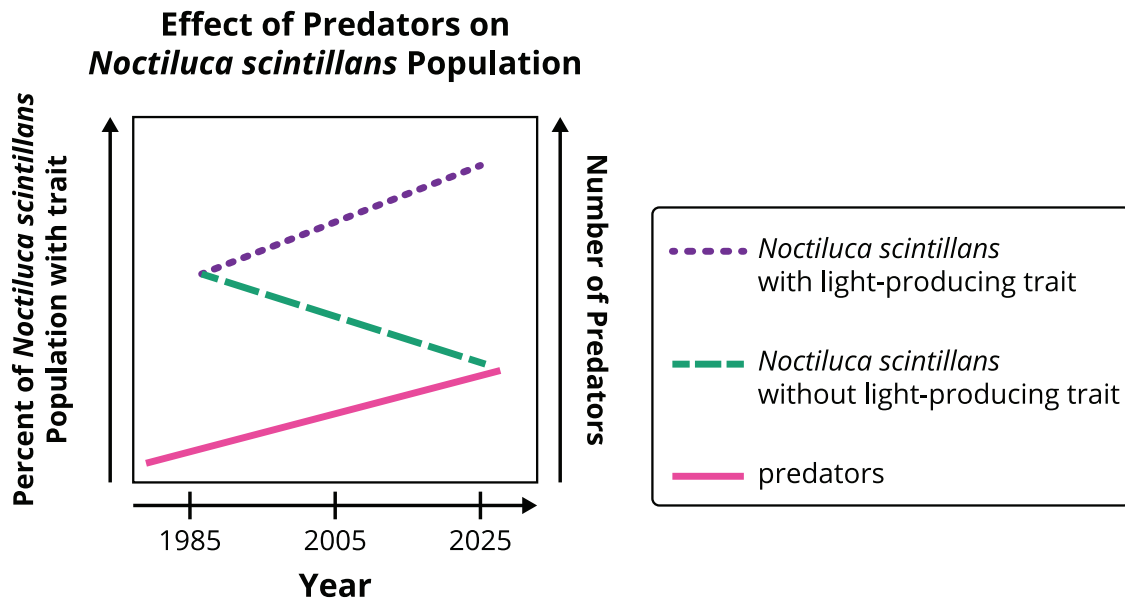
The *Noctiluca scintillans*' ability to produce light is a(n) (**advantageous**, **disadvantageous**) trait in Seawatch Bay, because individuals with that trait are (**less**, **more**) likely to survive and reproduce.

Review each question carefully. You may not return to a previous question.

5. Use the information to answer the question.

Seawatch Bay

The scientists collect data on how the number of *Noctiluca scintillans* in Seawatch Bay with the light-producing trait and without the light-producing trait has changed in relation to the predator population that eats them. The graph shows the data.



Based on the data, the scientists claim that the ability to produce light is an advantageous trait for most microorganisms.

A fellow researcher recommends that the scientists' claim should be revised based on limitations in the data.

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Why should the scientists' claim be revised?

Select **all** that apply.

- A. The data does not include information gathered over multiple years.
- B. The data does not include information on microorganisms outside of Seawatch Bay.
- C. The data does not include a description of how traits are expressed in populations.
- D. The data does not include the population change of *Noctiluca scintillans* that do not produce light as a control.
- E. The data does not include information from multiple microorganisms that produce light, only the *Noctiluca scintillans*.

Review each question carefully. You may not return to a previous question.

6. Use the information to answer the question.

Water Balloon Challenge

Teachers volunteer to get soaked for a water balloon challenge. Each student team gets one balloon to launch, and the goal is to get water on as much of the teacher as possible. The students can fill their balloons with as much water as they want, and they must use a catapult to launch the balloons. The catapult can launch the balloon at a maximum speed of 10 meters per second (m/s).

The teams need to figure out the best balloon mass and the best launch speed to get the most water on the teachers. The students measure the mass of five water balloons. They set the catapult to launch the balloons at a speed of 7 m/s. They launch each balloon and calculate its kinetic energy. The table shows the data.

Water Balloon Mass and Kinetic Energy

Balloon Number	Mass (kilograms)	Speed (meters per second)	Kinetic Energy (joules)
1	0.16	7	3.9
2	0.20	7	4.9
3	0.24	7	5.9
4	0.40	7	9.8
5	0.48	7	11.8

Which statement BEST describes the relationship shown in the data table?

Write the correct choice in each blank to complete the sentence.

mass speed doubles halves

When the _____ of the water balloon doubles, its kinetic energy _____.

Review each question carefully. You may not return to a previous question.

7. Use the information to answer the question.**Water Balloon Challenge**

The students decide to launch water balloons at different speeds to see how speed affects the kinetic energy of the balloons. The students use five water balloons that all have a mass of 0.6 kilograms. The table shows the data.

Water Balloon Speed and Kinetic Energy

Balloon Launch Speed (m/s)	Kinetic Energy (joules)
4	4.8
5	7.5
6	10.8
7	14.7
8	19.2

Which statement BEST identifies the cause and effect relationship between speed and kinetic energy?

- A. When kinetic energy increases, speed increases.
- B. When kinetic energy increases, speed decreases.
- C. When speed increases, kinetic energy increases.
- D. When speed increases, kinetic energy decreases.

Review each question carefully. You may not return to a previous question.

8. Use the graph to answer the question.

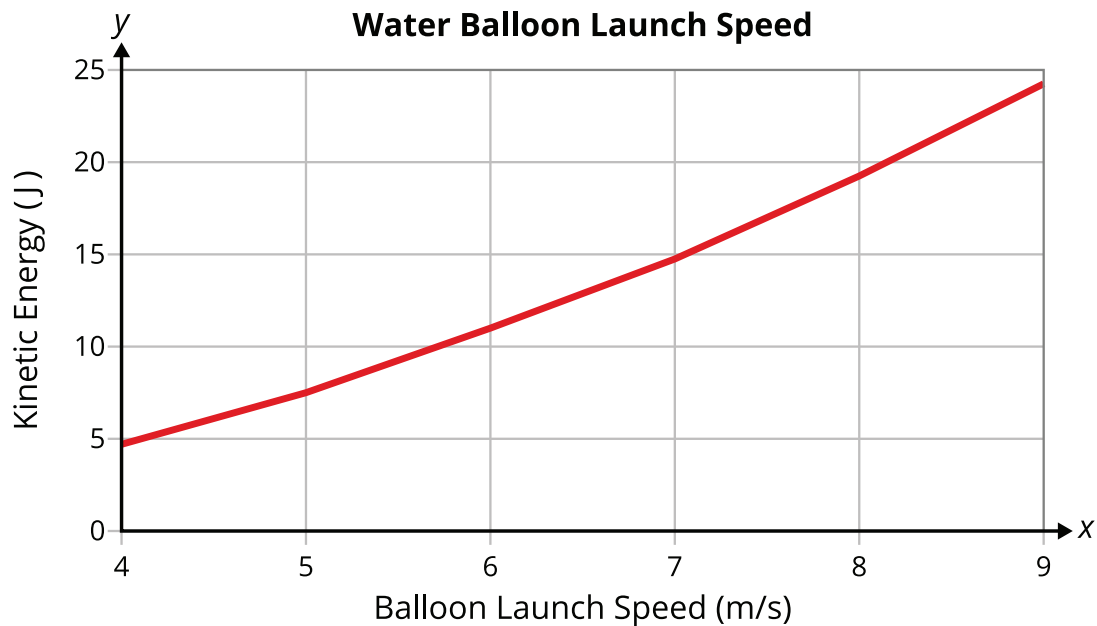
Water Balloon Challenge

The students decide to launch water balloons at different speeds to see how speed affects the kinetic energy of the balloons. The students use five water balloons that all have a mass of 0.6 kilograms. The table shows the data.

Water Balloon Speed and Kinetic Energy

Balloon Launch Speed (m/s)	Kinetic Energy (joules)
4	4.8
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7	14.7
8	19.2

The students graph the data they collected.



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What is the approximate kinetic energy, in joules (J), of a 0.6-kilogram water balloon moving at a speed of 9 m/s?

- A. 11 J
- B. 19 J
- C. 24 J
- D. 29 J

Review each question carefully. You may not return to a previous question.

9. Use the information to answer the question.

Water Balloon Challenge

To win the water balloon challenge, students need to maximize the mass of the water balloon and avoid having it burst before it can be catapulted.

Students fill 7 identical water balloons with water until they reach masses between 0.90 and 0.94 kilograms. They observe the water balloons to see if they immediately burst and record the data.

Maximum Water Balloon Mass

Mass (kg)	0.90	0.91	0.92	0.93	0.94	0.94	0.94
Does it burst?	No	No	Yes	No	Yes	Yes	Yes

A student claims that the maximum mass for the water balloon should be 0.91 kg. Another student is concerned that there may be a measurement error.

What method would help the students improve the accuracy of the data?

- A. Expand the range of masses from 0.85 kg to 0.99 kg.
- B. Fill each of the 7 water balloons with 0.90 kg of water instead of different masses.
- C. Measure the mass of the water balloons before they are filled with water to see if they can hold more than 0.91 kg of water.
- D. Conduct multiple trials using water balloons with masses between 0.91 kg and 0.94 kg to determine at what mass the water balloons burst.

Review each question carefully. You may not return to a previous question.

10. Use the information to complete the task.**Water Balloon Challenge**

Students have discovered this information.

- The mass and the speed of a water balloon affects its kinetic energy.
- The maximum mass for the water balloons is 0.92 kilograms.
- Water balloons with the most kinetic energy will have the biggest splash radius on impact.

The students want to make sure they are creating the largest splash radius possible. They lean a large piece of cardboard against a wall to act as a target. They catapult a 0.46-kg water balloon at the target at a speed of 8 m/s and mark the splash radius with a marker. Next, they catapult a 0.92-kg water balloon at the target at a speed of 8 m/s and mark its splash radius with a marker.

They confirm that the water balloon with the larger mass has a much larger splash radius than the water balloon with the smaller mass.

What can the students do to create an even larger splash radius?

Use the choices to complete the sentences.

lower

higher

decrease

increase

The splash radius will be even larger if the 0.92-kg balloon is catapulted at a _____ speed at the target.

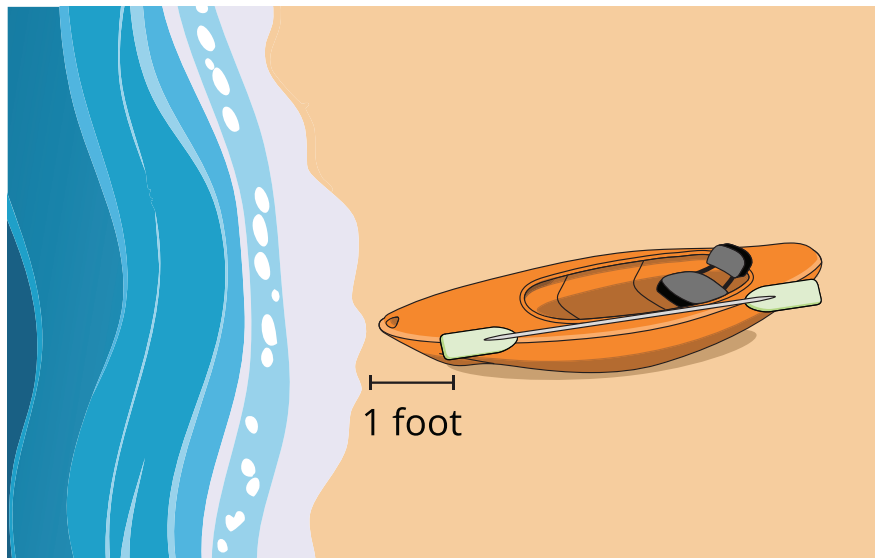
An increase in speed causes a(n) _____ in kinetic energy.

Review each question carefully. You may not return to a previous question.

11. Use the information to answer the question.

Runaway Kayak

Keri visited Santa Barbara, California. On May 10th, she went kayaking in an ocean bay with calm water. She left her kayak on dry sand about one foot away from the waterline at high tide. She checked on the kayak every few days to make sure it was still not touching the high tide waterline, and she made notes in her journal.

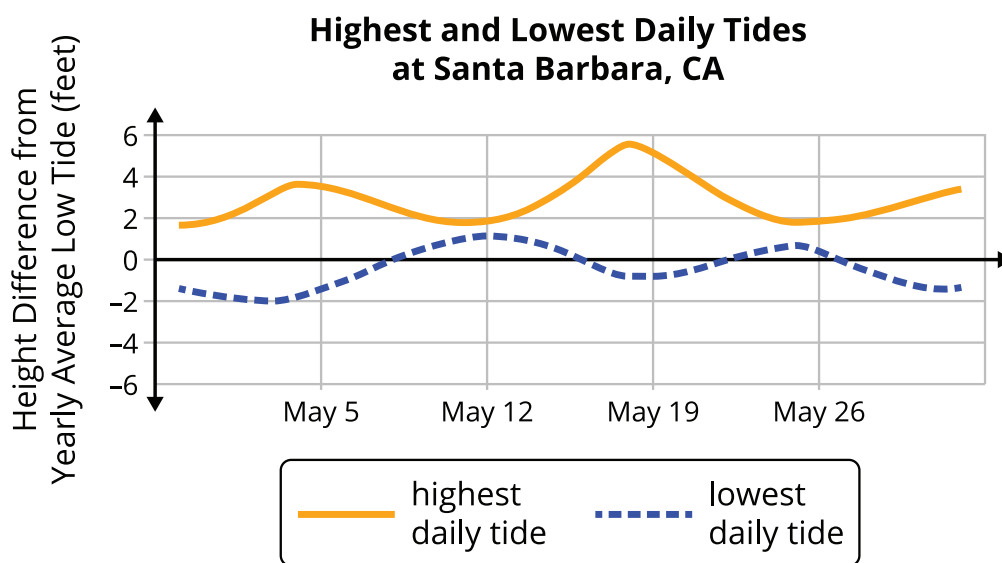


Date	Journal Entry
May 10 th	Went kayaking then pulled the kayak onto the beach.
May 12 th	Checked on the kayak. It is still on the beach not touching the waterline.
May 14 th	Had a picnic on the beach. The kayak is still on the dry sand.

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Keri decides to go kayaking again on May 18th. When she returns to the beach, her kayak is floating in the water. Keri wonders what happened.

She looks at this graph that shows the highest and lowest daily tides for May of this year in Santa Barbara.



What claim can be made based on the data?

Select the correct words to complete the sentence.

The kayak is floating in the water because the **(high, low)** tide on May 18th caused the water level at the beach to **(fall, rise)** more than at any other time during the month.

Review each question carefully. You may not return to a previous question.

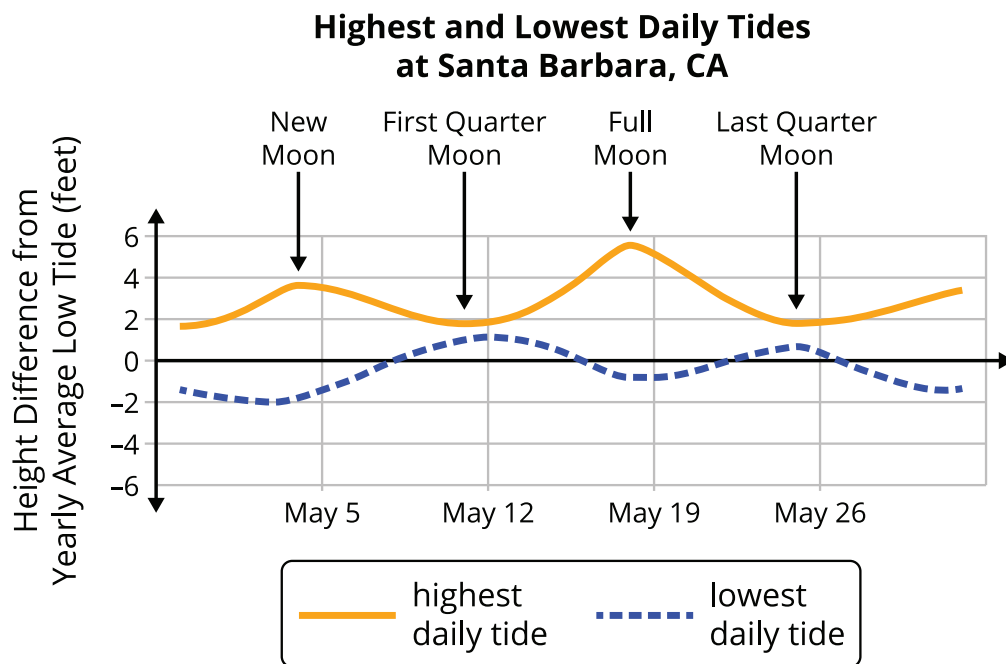
12. Use the information to answer the question.

Runaway Kayak

Keri visited Santa Barbara, California. On May 10th, she went kayaking in an ocean bay with calm water. She left her kayak on dry sand about one foot from the waterline at high tide. She checked on the kayak every few days to make sure it was still not touching the high tide waterline, and she made notes in her journal.

Keri decides to go kayaking again on May 18th. When she returns to the beach, her kayak is floating in the water.

During the high tide on May 18th, the water rose higher than earlier in the month. Keri learns that tides are affected by the phases of the moon. She finds information about the phases of the moon for May of this year and adds it to the tide graph.



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What phase is the moon in when Keri finds her kayak floating in the water?

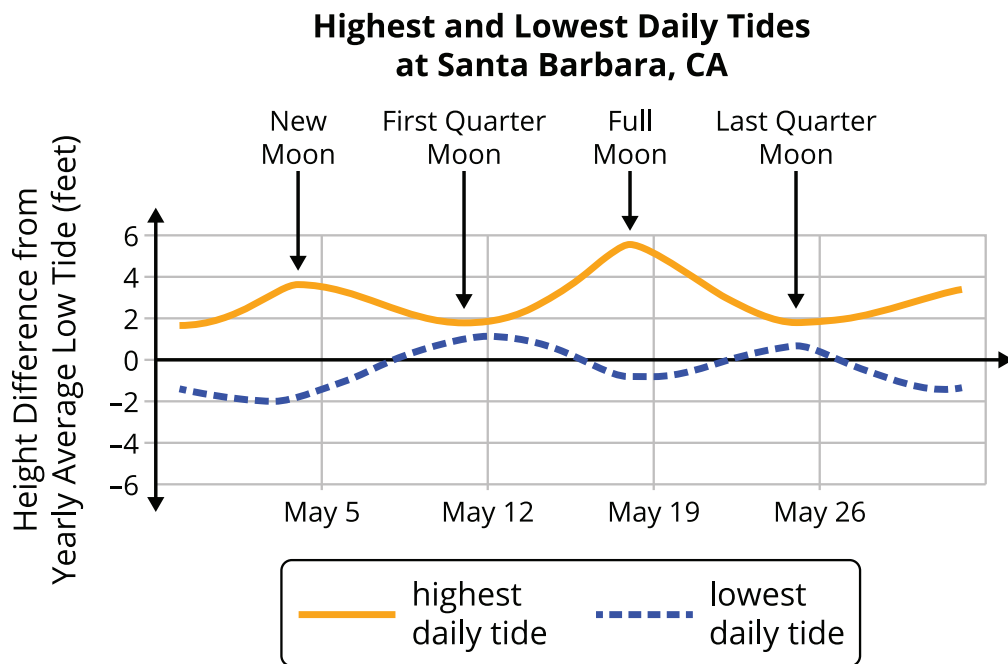
- A. full moon
- B. new moon
- C. last quarter moon
- D. first quarter moon

Review each question carefully. You may not return to a previous question.

13. Use the information to answer the question.

Runaway Kayak

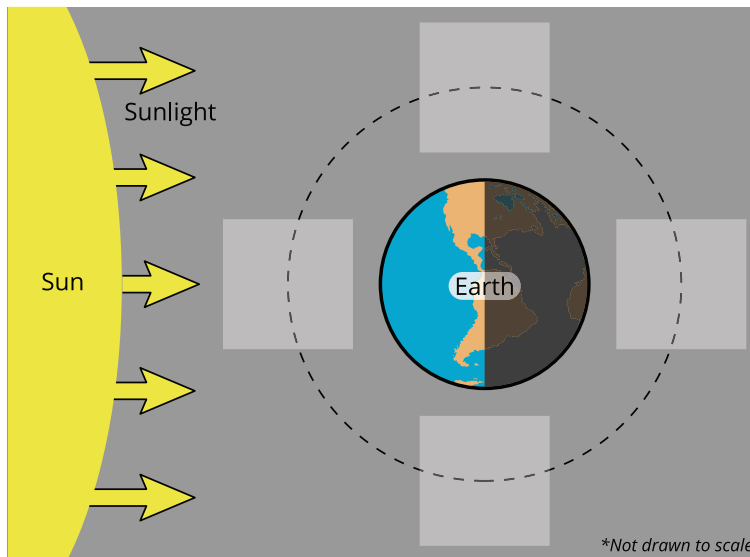
During the high tide on May 18th, the water rose higher than earlier in the month. Keri learns that tides are affected by the phases of the moon. She finds information about the phases of the moon for May of this year and adds it to the tide graph.



It was a full moon on May 18th when Keri found her kayak floating in the water. She wants to make a model to show where the Moon was relative to Earth on that day.

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Indicate the correct position of the Moon to show the relationship between the Sun, Earth, and Moon on May 18th.



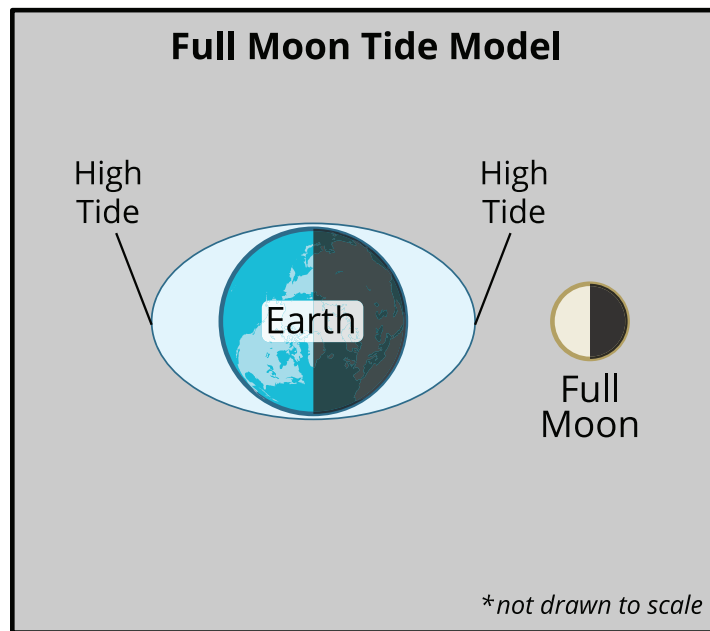
Review each question carefully. You may not return to a previous question.

14. Use the information to answer the question.

Runaway Kayak

Keri researches how tides are affected by the moon. She learns that the Moon's gravitational pull on Earth's oceans causes the water to be redistributed. This results in bulges of water, or high tides, on the sides of Earth closest to the Moon and farthest from the Moon.

She creates a new model to represent the information she finds.



Keri claims that high tides are higher when it is a full moon than when it is a first quarter moon or last quarter moon, because the Moon is in line with Earth and the Sun.

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How could Keri revise the model to better support her claim?

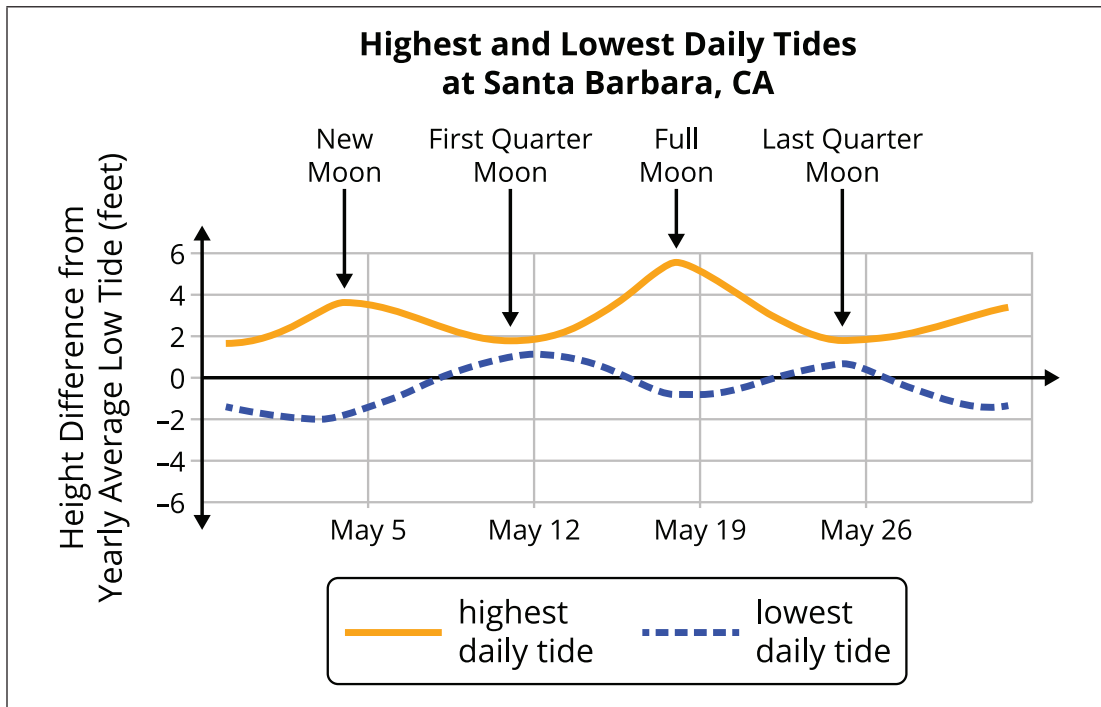
Select **all** that apply.

- A. Include the Sun in the model.
- B. Make Earth appear larger relative to the Moon.
- C. Add arrows to represent Earth rotating on its axis.
- D. Show how the bulge of water changes when it is not a full moon.
- E. Show how solar energy travels in a straight line from the Sun to Earth and the Moon.

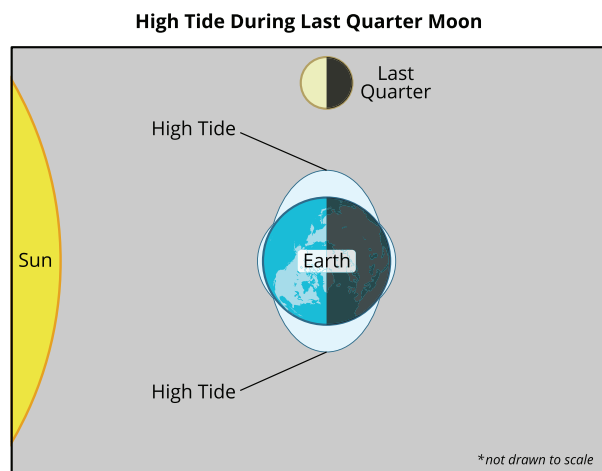
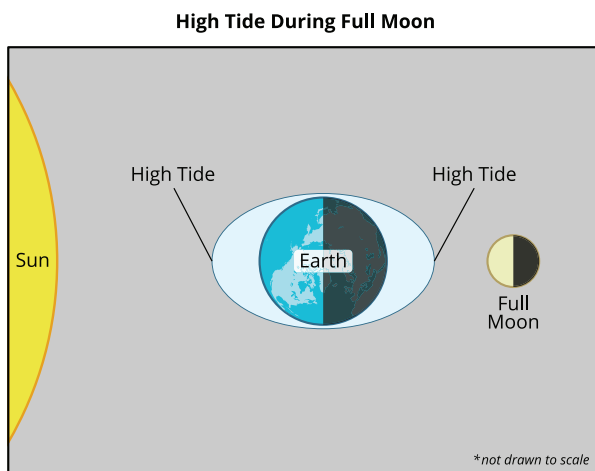
Review each question carefully. You may not return to a previous question.

15. Use the information to answer the question.

Runaway Kayak



The model that Keri revised shows the high tide during a full moon and during a last quarter moon.



Keri pulls her kayak out of the water on May 18th. She leaves it on the dry sand about one foot from the water line at high tide. She will not return to the beach until May 26th.

This question has two parts. Answer Part A, and then answer Part B.

Part A

Can Keri leave her kayak on the beach and expect it to not be in the water when she returns on May 26th?

Select the correct words to complete the sentences.

Keri (**can, cannot**) leave her kayak on the beach. The moon phase will change from a full moon to a (**new, first quarter, last quarter**) moon between May 18th and May 26th.

Part B

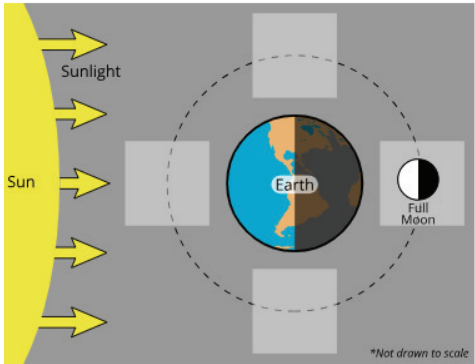
What reasoning BEST supports the answer to Part A?

Select the correct words to complete the sentences.

The high tide on May 26th will be (**higher, lower**) than it was during the full moon on May 18th when Keri's kayak floated in the water. This is because high tides are lower when the Moon (**is, is not**) in line with Earth and the Sun.

**NSCAS General Grade 8
Practice Test Answer Key
Science (English)**



Sequence	Key	Points
1	luciferase gene--> luciferase protein--> trait to produce light	1
2	A, E	1
3	C	1
4	Part A: B Part B: advantageous, more	2
5	B, E	1
6	mass, doubles	1
7	C	1
8	C	1
9	D	1
10	higher, increase	1
11	high, rise	1
12	A	1
13		1
14	A, D	1
15	Part A: can, last quarter Part B: lower, is not	2